llmXive Automated Review of Vision as Unified Multimodal Generation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a comprehensive evaluation of SenseNova-Vision across four vision task families. The claim-accuracy review focuses on whether the textual claims in the abstract, introduction, and results sections are strictly supported by the provided tables and citations.

1. Overstatement of "Leading" Status in Object Detection: In Section 5.1 and Table 1, the text states that SenseNova-Vision "leads on structured visual understanding" and specifically highlights the object detection result. Table 1 reports a score of 56.6 for SenseNova-Vision on COCO-Common, which is identical to the score of Grounding DINO-Swin-T (56.6). The verb "leads" typically implies a strict inequality (Score > Baseline). Since the scores are tied, the claim is technically inaccurate. The authors should either qualify the statement to "matches or leads" or verify if a secondary metric breaks the tie.

2. Ambiguity in Baseline Comparisons for Dense Geometry: Section 5.2 claims the model "outperforms recent generation-based baselines." Table 2 supports this against Marigold and DICEPTION. However, the table also includes FE2E (4.1 abs rel) and Lotus-2 (4.1 abs rel), which are generation-based and very close to SenseNova (4.0). The text does not explicitly distinguish between "generation-based" and "geometry-specialized" baselines in the narrative flow, potentially leading a reader to believe SenseNova outperforms *all* listed baselines, whereas it is only marginally better than FE2E and Lotus-2. The text should clarify that it outperforms *most* generation-based baselines or explicitly name the ones it beats.

3. Magnitude of Gap in Multi-View Geometry: In Section 5.4, the authors state SenseNova "approaches the leading specialist" (VGGT). While the 7Scenes F1 score (87.9 vs 88.4) is close, the ETH3D F1 score shows a notable gap: SenseNova (72.2) vs VGGT (80.9). A ~10% relative drop on a major benchmark like ETH3D is significant. The claim "approaches" is defensible but risks over-optimism without acknowledging the specific deficit on ETH3D. The text should be tempered to reflect that it approaches specialists on some metrics but lags on others (specifically ETH3D reconstruction).

4. Segmentation Results Specificity: In Section 5.3, the claim of "strong results on reasoning segmentation" is supported by the Val set (63.2 vs 62.1 for LENS). However, the Test set shows a smaller margin (60.7 vs 57.2). While the claim holds, the text should ensure it does not imply dominance across all splits if the margin is narrow or if the Test set performance is the primary benchmark for the field.

Overall, the paper's core claims are largely supported by the data, but several instances of "leading" or "outperforms" are slightly too strong given the specific numbers in the tables (ties or narrow margins). These are minor textual adjustments rather than scientific flaws.

Action items.

- [**writing**] The paper presents a comprehensive evaluation of SenseNova-Vision across four vision task families. The claim-accuracy review focuses on whether the textual claims in the abstract, introduction, and results sections are strictly supported by the provided tables and citations. 1. Overstatement of "Leading" Status in Object Detection: In Section 5.1 and Table 1, the text states that SenseNova-Vision "leads on structured visual understanding" and specifically highlights the object detection result.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure provides a clear visual overview of the model's capabilities, but the caption contains a file reference mismatch and uses terminology that does not align with the specific labels used in the image panels.

Figure 2

The figure serves as a high-level system overview but fails to visually demonstrate the specific 'annotation conversion' process described in the caption, relying instead on a list of training design features and generic input examples.

Figure 3

Figure 3 provides a broad overview of task outputs but lacks colorbars for depth/normal maps and omits scale/coordinate information for 3D reconstructions, limiting interpretability.

Figure 4

The figure effectively visualizes the composition of the SN-VC source data, but the x-axis label 'Configured examples' is confusing for a count-based chart, and the sub-category text is difficult to read.

Figure 5

Figure 5 effectively demonstrates the model's diverse capabilities across multiple tasks, but the legend's color coding does not perfectly align with the section headers, and the depth map visualization lacks a necessary scale bar.

Figure 6

The figure demonstrates the segmentation capability but fails to visually link the text-encoded point cues to the resulting masks, as the coordinate labels are placed in the corners rather than near the indicated targets.

Figure 7

The figure suffers from significant rendering issues, including raw HTML code in the header and a misleading 'Original' panel that claims to show annotations but displays a plain photo. Additionally, the X-SAM result fails to perform instance segmentation, showing only a single merged mask.

Figure 8

The figure demonstrates instance segmentation but appears to miss the white onion on the second shelf, and the caption lacks sufficient detail to explain the specific input prompt used.

Figure 9

The figure demonstrates a segmentation result but fails to match its caption's claim of showing 'both examples' by displaying only one case. Additionally, the image includes a descriptive text block defining color codes that is absent from the caption, reducing the figure's self-containment.

Figure 10

The figure fails to demonstrate word-level text segmentation as claimed; instead of showing a mask overlay, it generates the text string itself on a black background. Additionally, the figure lacks sufficient descriptive context in the caption to understand the specific task execution.

Figure 11

Figure 11 is a clear and well-structured pipeline diagram that effectively visualizes the four data construction engines described in the caption. All steps, inputs, and outputs are legible, and the flow is logical without clutter or missing information.

Figure 12

The figure provides a broad overview of structured visual understanding tasks, but the 'Pointing' row is compromised by a missing color scale for the dense grid and illegible text annotations in the cat example.

Action items.

- [**writing**] Figure 1: The caption states the figure shows 'dense geometric prediction' and 'multi-view visual geometry', but the image labels use different terminology ('Surface Normal', 'Depth', '3D Reconstruction & Camera Pose Estimation'). The caption should be updated to match the specific task labels shown in the figure.
- [**writing**] Figure 1: The file reference in the caption '[fig2_qualitative_overview.pdf]' contradicts the figure number (Figure 1). This should be corrected to match the actual figure index.
- [**writing**] Figure 2: The caption states the figure shows 'heterogeneous computer vision

annotations' being converted into generation targets, but the visual content is a high-level marketing overview (showing 'Training Design' and 'Instruction Input') rather than a technical diagram of the annotation-to-target conversion pipeline described.

- [**writing**] Figure 2: The 'Instruction Input' section lists specific tasks (e.g., 'OCR', 'bounding box') with circled numbers (1-13) that correspond to the top panels, but the text itself is not formatted as a code block or distinct data sample, making the 'native text' target format ambiguous.
- [**writing**] Figure 3: The caption refers to 'Figure 3' but the file name is 'fig4_data_examples.pdf', suggesting a mismatch between the figure label and the source file.
- [**science**] Figure 3: The 'Depth' and 'Surface Normal' outputs are color-coded but lack a colorbar or legend to explain the mapping between colors and values.
- [**science**] Figure 3: The '3D Reconstruction' output shows 3D point clouds but does not specify the coordinate system or scale, making quantitative interpretation impossible.
- [**science**] Figure 4: The x-axis label 'Configured examples' is semantically incorrect for a bar chart displaying dataset composition counts (labeled in millions, e.g., '46.7M'); the label should likely be 'Number of examples' or 'Dataset size'.
- [**writing**] Figure 4: The sub-category breakdowns (e.g., 'Point maps 24.9M / Cam. pose 21.8M') are rendered in a font size that is significantly smaller than the main labels and axis, reducing legibility.
- [**writing**] Figure 5: The bottom legend uses colored squares to define categories (Structured 2D, Dense Geometry, Segmentation, Multi-view 3D), but the corresponding section headers in the grid use different background colors (e.g., 'Depth' has a purple header while the legend maps 'Dense Geometry' to a light purple square), creating a visual disconnect between the legend and the figure structure.
- [**writing**] Figure 5: The 'Depth' section contains a depth map visualization but lacks a colorbar or scale legend to interpret the depth values represented by the colors.
- [**science**] Figure 6: The 'point' labels (e.g., [0.186, 0.081]) are positioned in the top-left corner of the images, far from the actual targets indicated by the blue masks (e.g., the cat, the church), making the spatial correspondence between the coordinate cue and the segmentation result unclear.
- [**writing**] Figure 6: The 'point' labels are rendered in a very small, low-contrast font that is difficult to read against the image backgrounds.
- [**science**] Figure 7: The 'Original' panel is labeled '(134 instances annotated)' but displays a standard RGB photograph with no visible annotations, masks, or instance IDs, making the claim unverifiable and the panel misleading as a ground truth reference.
- [**science**] Figure 7: The 'X-SAM' panel shows a single large blue blob rather than distinct instance masks, failing to demonstrate 'instance segmentation' for the crowded scene and contradicting the figure's purpose.

- [**writing**] Figure 7: The top text block contains raw HTML tags (e.g., <p>, <color>) and prompt instructions rather than a clean figure title or description, indicating a rendering or formatting error.
- [**science**] Figure 8: The segmentation mask on the right fails to include the white onion on the second shelf, which is a distinct object visually similar to the ketchup bottles, suggesting incomplete instance detection.
- [**writing**] Figure 8: The caption 'VGD segmentation with point reference cue' is too brief to explain the input prompt or the specific task context shown in the 'Original' image.
- [**science**] Figure 9: The caption claims to show 'Both examples' of free-form color-coded mask generation, but the rendered image displays only a single case (Original vs. SenseNova-Vision), creating a discrepancy between the text and the visual content.
- [**writing**] Figure 9: The image contains a large text block at the top describing specific color mappings (green for cat, blue for suitcase-1, etc.) that are not present in the provided caption, making the figure's context dependent on external text rather than self-contained.
- [**science**] Figure 10: The output image displays the text 'Coke' on a black background rather than a segmentation mask (e.g., a binary mask or overlay) on the original image, failing to demonstrate the 'segmentation' task described in the caption.
- [**writing**] Figure 10: The figure lacks a descriptive caption explaining the input prompt, the specific bottles shown, or the expected output format, relying solely on the generic title 'Word-level text segmentation'.
- [**science**] Figure 12: The 'Pointing' row contains a dense grid of colored dots (resembling a heatmap) but lacks a colorbar or legend to define the value scale or meaning of the colors.
- [**writing**] Figure 12: The 'Pointing' row includes a cat image with a purple text box that is illegible due to low resolution and poor contrast.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-written and avoids excessive in-group slang, but there are a few instances where acronyms and specific metric notations are introduced without sufficient immediate context for a competent reader from an adjacent field.

First, in Section 3.1, the acronym "SN-VC" is introduced. While the full name "SenseNova-Vision Corpus" is provided in the same sentence, the acronym is then used repeatedly in the subsequent paragraphs. It is best practice to ensure the full name is clearly established before the acronym is used, or to ensure the definition is prominent. The current usage is borderline but acceptable if the reader is assumed to have read the abstract; however, for a strictly self-contained body text, a clearer initial definition is preferred.

Second, Section 4.1 mentions the "rectified-flow training objective." While "rectified flow" is a known technique in the generative modeling community, the specific phrasing and its application here as a named objective for this model might be opaque to a reader from a neighboring field (e.g., standard computer vision or NLP) who is not deeply versed in the latest diffusion flow literature. A brief parenthetical explanation or a citation to the foundational paper would improve accessibility.

Third, the metrics "F1@mIoU" and "F1@Point" in Section 5.1 and Table 1 are used without explicit definition of the threshold parameters (e.g., IoU threshold). While "mIoU" is standard, the "F1@" notation implies a specific calculation (F1 score at a specific threshold) that is not universally standardized across all vision subfields. Defining the threshold (e.g., "at IoU=0.5") in the table caption or text would remove ambiguity.

Finally, the acronyms "RRA" and "RTA" in Section 5.4 are defined, but the specific criteria for "accuracy" (the 30-degree threshold) is mentioned in the same sentence. This is acceptable, but ensuring the definition is tight and clear is important.

Overall, these are minor issues that can be resolved with small textual additions to ensure the paper is fully self-contained for a broad technical audience.

Action items.

- [**writing**] Section 3.1 introduces 'SN-VC' without explicitly defining the acronym at first use in the body text. Add '(SN-VC)' immediately after 'SenseNova-Vision Corpus' in the first sentence of Section 3.1 to ensure self-containment.
- [**writing**] Section 4.1 uses 'rectified-flow training objective' without a brief gloss or citation. Add a parenthetical reference (e.g., 'following rectified flow [Citation]') or a one-clause definition to clarify the specific method for adjacent-field readers.
- [**writing**] Section 5.1 and Table 1 use 'F1@mIoU' and 'F1@Point' without defining the IoU threshold. Specify the threshold (e.g., 'at IoU=0.5') in the table caption or text to ensure the metric is operationally clear.
- [**writing**] Section 5.4 introduces 'RRA' and 'RTA' with definitions, but the 30-degree threshold for accuracy is buried in the sentence. Tighten the definition to explicitly state 'accuracy within a 30-degree threshold' immediately after the acronym expansion.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a logically coherent argument for unifying computer vision tasks under a single multimodal generation framework. The central thesis—that heterogeneous visual outputs (text, images, mixed) can be natively handled by a Unified Multimodal Model (UMM) without task-specific heads—is consistently supported by the proposed methodology and experimental results.

The argument structure is sound:

1. **Premise:** Current vision systems are fragmented by task-specific architectures.
2. **Proposal:** Cast all tasks as generation targets (text for symbols, images for dense maps) within a UMM.
3. **Method:** Construct a corpus (SN-VC) converting annotations to these native formats and fine-tune a base UMM (Bagel).
4. **Evidence:** Tables 1-4 demonstrate competitive performance across four distinct families (structured understanding, dense geometry, segmentation, multi-view) using the unified interface.
5. **Conclusion:** The unified approach is a scalable route for integrating vision into foundation models.

There are no internal contradictions between sections. The definitions of task families in Section 3 (Data) align perfectly with the evaluation metrics and results presented in Section 5 (Experiments). The distinction between text-based outputs (detection, OCR) and image-based outputs (depth, masks) is maintained consistently throughout the text and tables. The claim in the Abstract and Conclusion that the model "matches leading task-specialized systems" is supported by the data in Tables 1-4, where SenseNova-Vision achieves state-of-the-art or near-state-of-the-art results on multiple benchmarks without contradicting the limitations or specific performance gaps noted in the text (e.g., the gap in multi-view reconstruction compared to specialized geometry models is acknowledged in Section 5.4).

The logical flow from the "Data Protocol" to the "Training" strategy and finally to the "Experiments" is tight. The ablation of "task-specific heads" is a core premise, and the results consistently reinforce that the model achieves these results *without* them, as the architecture remains that of the base UMM. No non-sequiturs or unsupported causal leaps were found; causal language (e.g., "enables," "allows") is appropriately grounded in the described mechanism of unified training. The paper is internally consistent.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes a strong case for unifying computer vision tasks under a multimodal generation framework, but the rhetoric in the abstract and introduction occasionally exceeds the scope of the quantitative evidence provided.

Specifically, the abstract claims the model “matches leading task-specialized systems” across the board. However, a close reading of the results tables reveals a more nuanced picture. In Table 3 (Dense Geometric Prediction), SenseNova-Vision is outperformed by specialized models like MoGe-2 and DepthAnything V2 on several key metrics (e.g., NYUv2 abs rel, KITTI abs rel). Similarly, in Table 4 (Segmentation), it trails PSALM and X-SAM on generic and referring segmentation metrics. The claim of “matching” implies parity across the board, which the data does not support; the model is competitive in some areas but clearly behind in others. The text should be adjusted to reflect this variance, perhaps by stating it “achieves competitive performance” or “approaches” specialized systems, rather than a blanket “matches.”

Furthermore, the introduction states the model “leads on structured visual understanding.” While Table 1 shows strong results, it is not a universal lead; for instance, on the LVIS and Dense200 benchmarks, LocateAnything outperforms SenseNova-Vision. The claim should be qualified to reflect that it leads on *some* benchmarks or achieves state-of-the-art results on *specific* tasks, rather than implying a dominant lead across the entire family.

Finally, the conclusion asserts that the model “supports language-defined task variants beyond the training set.” This is an exciting finding, but the evidence is entirely qualitative (Section 5.3.4, Figures 8-10). While the qualitative examples are illustrative, the conclusion should be careful not to present this as a fully validated, general capability without quantitative backing. Phrasing it as “qualitative results suggest the potential to support...” would be more accurate and honest about the current scope of the evidence.

These are primarily issues of rhetorical precision rather than fundamental flaws in the science. The results are impressive, but the claims should be tightened to match the specific boundaries of the data presented.

Action items.

- [**writing**] The paper makes a strong case for unifying computer vision tasks under a multi-modal generation framework, but the rhetoric in the abstract and introduction occasionally exceeds the scope of the quantitative evidence provided. Specifically, the abstract claims the model “matches leading task-specialized systems” across the board. However, a close reading of the results tables reveals a more nuanced picture. In Table 3 (Dense Geometric Prediction), SenseNova-Vision is outperformed by specialized m

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a large-scale computer vision model trained on a corpus constructed from public datasets and generated annotations. A review of the data construction section (Section 3) and the appendix reveals that the authors explicitly state the corpus is built from “public images” and “available public annotations” (Section 3.2). The released artifact, SN-VC-50M, is described as containing “generated and curated targets” and “source lists, prompt templates, conversion rules,” while explicitly noting that “raw RGB images from public datasets” are not redistributed to avoid licensing issues (Appendix, SN-VC-50M Release Summary).

The paper does not appear to use private human-subjects data, PII, or data scraped in violation of Terms of Service. The datasets listed in the appendix (e.g., COCO, OpenImages, Cityscapes, ScanNet) are standard public benchmarks with established licenses for research use. The use of synthetic data generation tools (e.g., MoGe-2, LingBot-Depth) to create dense labels from sparse or missing annotations is a standard practice in the field and does not introduce new ethical risks regarding consent or privacy, provided the source images are public, which the authors assert.

There is no evidence of dual-use capabilities described in a manner that lowers the barrier to specific harms (e.g., automated surveillance of individuals, generation of deceptive media for disinformation) without mitigation. The model’s capabilities (detection, segmentation, depth estimation) are standard computer vision tasks. The paper does not disclose any operational vulnerabilities in live systems, nor does it involve human subjects requiring IRB approval.

Consequently, no specific safety or ethics risks requiring mitigation or disclosure were identified. The paper adheres to standard norms for data provenance and release in the computer vision community.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling unified framework for computer vision tasks, but the evidentiary strength of the quantitative claims is currently weakened by a lack of reported variance and potential confounds in baseline comparisons.

First, the primary quantitative results in Tables 1 through 4 are presented as single-point estimates (e.g., “56.6” mAP, “4.0” abs rel) with no accompanying standard deviation, confidence intervals, or indication of the number of random seeds used. In modern deep learning benchmarks, especially with large models, performance can fluctuate significantly across seeds. A claimed improvement of 0.5 to 1.0 points over a strong baseline (as seen in several entries) is often indistinguishable from sampling noise without variance reporting. To support the claim that SenseNova-Vision “matches” or “leads” these systems, the authors must demonstrate that these gains are stable across re-initializations. Reporting mean $\pm$ std over at least 3-5 seeds is the standard requirement to rule out luck.

Second, there is a risk of unfair baseline tuning asymmetry. Section 4 details the specific training setup for SenseNova-Vision (50K steps, specific LR, warmup), but the baselines are largely cited from their original papers. It is unclear if the strong task-specialized baselines (e.g., DepthAnything V2, Grounding DINO) were re-tuned with the same hyperparameter search budget and compute resources as the proposed method. If the proposed method benefited from extensive tuning while baselines were used with their default or sub-optimal settings, the reported "improvement" may reflect the tuning effort rather than the architectural unification. The authors should either disclose that baselines were re-tuned to match the search effort or provide a controlled ablation where the proposed method is compared against baselines with matched training budgets.

Finally, regarding the "re-evaluated" baselines in Table 2 (marked with *), the text notes that MoGe-2 was re-evaluated for consistency. However, it is ambiguous whether* all* geometry-specialized baselines were re-run under the exact same inference protocol (resolution, preprocessing, decoding logic) as the proposed method. If some baselines were taken from original papers with different inference settings, the comparison is confounded. Explicit confirmation that all baselines were re-run with identical inference protocols is necessary to validate the fairness of the comparison.

Addressing these points—specifically by adding seed variance and clarifying baseline tuning/inference protocols—will significantly strengthen the scientific evidence supporting the paper's central claims.

Action items.

- [**writing**] The paper presents a compelling unified framework for computer vision tasks, but the evidentiary strength of the quantitative claims is currently weakened by a lack of reported variance and potential confounds in baseline comparisons. First, the primary quantitative results in Tables 1 through 4 are presented as single-point estimates (e.g., "56.6" mAP, "4.0" abs rel) with no accompanying standard deviation, confidence intervals, or indication of the number of random seeds used. In modern deep l

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in the experimental section (Section 5) is insufficient for rigorous quantitative claims. While the paper presents extensive benchmark results across four task families, it consistently reports single point estimates (e.g., "56.6" mAP in Table 1, "4.0" abs rel in Table 2) without any accompanying measure of uncertainty such as standard deviation (SD), standard error (SE), or confidence intervals (CI).

In the context of deep learning, where training involves stochastic optimization and random data sampling, performance metrics naturally vary across different random seeds. Reporting a single number from one run (or an unreported average) makes it impossible to determine if the observed differences between SenseNova-Vision and baselines are statistically significant or merely artifacts of random initialization. For instance, in Table 1, SenseNova-Vision achieves 56.6 mAP on COCO-Common, while Grounding DINO-Swin-T achieves 56.6 as well (tied) and LocateAnything achieves 54.7. Without variance, we cannot assess the stability of these rankings.

Furthermore, the paper frequently uses language implying statistical superiority (e.g., "SenseNova-Vision achieves strong overall performance," "outperforming recent generation-based baselines") without providing the statistical machinery to support these assertions. The bolding of best results in tables is a common convention but does not constitute a statistical test. When comparing against multiple baselines across multiple datasets (e.g., 5 depth datasets in Table 2), the risk of false positives increases, yet no multiple-comparison correction (such as Bonferroni or Benjamini-Hochberg) is mentioned or applied.

To rectify this, the authors should re-run the experiments for the proposed method and key baselines with at least 3 independent random seeds. The results should be reported as mean ± SD. If the variance is small enough to support the claims, formal hypothesis tests (e.g., paired t-tests) should be conducted to validate the "outperforming" claims. If re-running is not feasible, the text must be softened to reflect that these are single-run observations, and the implication of statistical significance must be removed.

Action items.

- [**science**] Tables 1-4 report single point estimates (e.g., '56.6' mAP) for SenseNova-Vision and baselines without any measure of uncertainty (SD, SE, or CI). In deep learning, results vary across random seeds. Report mean ± SD over at least 3 independent training runs for the proposed method and key baselines to distinguish stable improvements from run-to-run noise.
- [**writing**] Section 5.1 claims SenseNova-Vision is 'strong' or 'competitive' based on point estimates. Without reported variance or formal hypothesis tests (e.g., paired t-tests or bootstrap tests across seeds), these claims of superiority are statistically unsupported. Add uncertainty metrics or explicitly state that comparisons are based on single-run point estimates.
- [**science**] Table 2 and Table 4 compare the proposed method against multiple baselines across several datasets (e.g., 5 depth benchmarks, 3 normal benchmarks). No correction for multiple comparisons (e.g., Bonferroni, Holm, or FDR) is applied to the 'significant' claims implied by the bolding of best results. Apply a correction or rephrase claims to avoid implying statistical significance where none was tested.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high standard of writing quality, allowing a reader to move through the argument with minimal friction. The abstract effectively summarizes the core formulation, the corpus construction, and the headline results, setting a clear expectation for the body of the paper.

The introduction is particularly well-structured. It opens with a clear motivation regarding the fragmentation of current vision systems, transitions smoothly into the limitations of prior unification attempts, and culminates in a precise statement of the proposed formulation. The topic sentences in the subsequent paragraphs are strong, immediately signaling the paragraph's focus (e.g., "Making this formulation trainable at scale requires...").

Throughout the Data and Training sections, the prose maintains a consistent technical voice and tense. The authors successfully navigate complex methodological details—such as the conversion of heterogeneous annotations into text, image, and mixed targets—without burying the main points. Paragraphs are logically grouped by task family (structured understanding, dense geometry, etc.), and each paragraph within these sections adheres to a single point, avoiding the common pitfall of mixing setup, results, and limitations in one block.

Transitions between sections are handled effectively. For instance, the move from the Data section to the Training section is bridged by a clear explanation of how the constructed corpus is utilized, rather than an abrupt shift in topic. The Experiments section is similarly well-organized, with clear lead-in sentences for each task family that orient the reader to the specific benchmarks and metrics being discussed.

The writing is concise and free of significant redundancy. While the paper is dense with technical information, the sentence structures are generally direct and parseable on the first pass. There are no instances of ambiguous pronoun references that would force a reader to backtrack, and the terminology (e.g., "SenseNova-Vision Corpus," "unified multimodal generation") is used consistently throughout. The conclusion effectively synthesizes the contributions without introducing new, unexplained concepts.

Overall, the paper is a model of clear scientific communication. The prose serves the science well, ensuring that the reader's cognitive load is focused on understanding the novel contributions rather than deciphering the text. No revisions are required to improve the writing quality.